

AADITYA RAJ

Ranchi, Jharkhand, India

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EDUCATION

Birla Institute of Technology, Mesra

Bachelor of Technology (Computer Science and Engineering) CGPA - 8.28

August 2023 – Present

Ranchi, Jharkhand

Delhi Public School, Ranchi

12th - 96.6%

2021 – 2023

Ranchi, Jharkhand

TECHNICAL SKILLS

Languages: Python, C/C++, SQL

Frameworks: Tensorflow, PyTorch, Keras, Huggingface, Langchain

Developer Tools: Git, Docker, Jupyter-Notebook, VS Code, PyCharm

Libraries: Pandas, NumPy, Matplotlib

EXPERIENCE

Research Assistant

Birla Institute of Technology, Mesra

January 2025 – Present

Ranchi, Jharkhand

- Working under a professor to replicate and enhance the findings of the paper *Powe of Alignment: Exploring the effect of face alignment on ASD Diagnosis using facial images*
- Investigating the impact of facial alignment on the accuracy of ASD diagnosis using deep learning models trained on facial image datasets.
- Fine-tuning state-of-the-art CNN architectures such as ResNet50, MobileNet, and Xception to evaluate performance changes due to alignment techniques.
- Experimenting with various face detection and preprocessing pipelines to optimize diagnostic accuracy and model generalization.

PROJECTS

CaptionFlow | *TensorFlow, Keras, OpenCV, Streamlit*

GitHub

- Developed a deep learning model for image captioning using a CNN + Transformer architecture.
- Processed over 300,000 image-caption pairs from the COCO2017 dataset.
- Deployed via Streamlit for real-time image captioning.
- Optimized inference with pre-trained embeddings and attention mechanisms.
- Improved accessibility and content annotation through automated caption generation.

ACHIEVEMENTS

- 2nd Rank** in Hack & Forge 2024 (**100+ teams**), Society for Data Science, BIT Mesra.

CERTIFICATIONS

- Coursera Deep Learning Specialization by Andrew Ng

View Certificate